

Qianqian Zhang

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EDUCATION

University of California, San Diego

Expected Jun. 2027

Major: Data Science | **Degree:** Master of Science

Current Courses: Recommender Sys&Web Mining, Data Science Programming, Numerical Linear Algebra, Probs. & Stats.

Wuhan University (WHU), Wuhan, China

Sep. 2021 – Jun. 2025

Major: Geographic Information System | **Degree:** Bachelor of Engineering | **GPA:** 3.79/4.0

Core Courses: Artificial Intelligence and Machine Learning (A), Engineering Information and Database System (A), Data Structure and Algorithms (A), Computer Vision and Pattern Recognition (A), Software Engineering (A)

TECHNICAL SKILLS

- **Job Interests:** interested in applying AI techniques and data science to address practical challenges in industrial applications.
- **Domains:** Recommendation System, Search Engine, LLM, CV, Spatial Data Analysis, Data Management and Visualization
- **Languages:** Python, C++, SQL, R
- **Framework:** PyTorch, TensorFlow
- **Distributed Systems:** Hadoop, Spark

WORK EXPERIENCE

Machine Learning Engineer Intern, Baidu (China) Co., Ltd., Shanghai, China

Jun. – Aug. 2024

- Developed and deployed an pretrained model for financial credit risk assessment, enhancing model's predictive performance with KS (21.58%→23.47%) and AUC (0.650→0.661).
- Built ~10,000-dimensional user feature vectors by integrating BAIDU user profiles, statistical and behavioral features.
- Applied a Progressive Layered Extraction (PLE) architecture to efficiently capture common representations while preserving task-specific parameters, leading to scalable and robust modeling of high-dimensional financial data.
- Applied model pruning to remove underperforming parameters, achieving ~17% reduction in model size.

SELECTED PROJECT

UnifiedSSR+: An Enhanced Framework for Unified Sequential Search and Recommendation

Oct.2024 – Jun.2025

(Under review at *ACM Transactions on Intelligent Systems and Technology*)

- Proposed a new Unified framework for Sequential Search, Recommendation and Query Prediction to jointly model cross-scenario and cross-view behaviors.
- Designed an Intent-oriented Session Modeling module with self-supervised signals to capture dynamic user intent through intra-sequence discrimination and cross-sequence alignment.
- Incorporated Multi-head Target Attention to integrate long- and short-term user intent, and applied Causal Attention across all layers to enforce unidirectional temporal consistency.
- Conducted extensive experiments on three public datasets and consistently achieved superior performance over most baselines across three scenarios.

Amazon KDD Cup 2024 Multi-Task Online Shopping Challenge for LLMs

Mar. – May. 2024

- Applied data augmentation and class distribution rebalancing techniques to improve dataset representativeness and mitigate the seesaw effect in multi-task optimization.
- Conducted parameter-efficient fine-tuning of the LLaMA-8B backbone using Low-Rank Adaptation (LoRA), enabling efficient domain adaptation under computational constraints.
- Implemented Multi-Task Alternating Training to address catastrophic forgetting and stabilize transfer in multi-task learning.
- Led an undergraduate team to advance to Phase 2, achieving a top 20% ranking in Phase 1.

Intelligent Driving Behavior and Pattern Analysis Platform Based on Trajectory Data

Sep. – Oct. 2024

- Preprocessed and cleaned 1.1M+ GPS trajectory records from private cars in Shenzhen to enable large-scale spatial data analysis.
- Constructed user representations by extracting spatiotemporal trajectory features and inferring five personality dimensions, supporting user profiling and personalized applications.
- Built an insurance pricing model based on driving behavior, demonstrating the application of spatial data analysis in the automotive industry.

Movie Recommendation System

Jan. – Mar. 2024

- Developed the re-ranking module of a Movie Recommendation System using the MovieLens dataset.
- Applied reinforcement learning strategies, leveraging the Thompson Sampling algorithm from multi-armed bandits to optimize the exploration – exploitation trade-off in candidate movie ranking.